Machine Vision Assignment 3

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ABSTRACT:
In this assignment, we applied morphological filtering, median filtering, and Laplacian filtering using MATLAB to process digital images. First, morphological operations are used for removing noise from a binary image while preserving its shape. Next, we applied both morphological and median filters to a fingerprint image, and then we compared their performance in noise removal. Finally, Laplacian filtering was used to enhance the edges of a grayscale moon image. The results showed that morphological filtering was effective for binary image processing, while median filtering preserved image details during noise removal. Laplacian filtering successfully sharpened the image by enhancing edges and fine surface details.

TEST RESULTS:

Fig1: Morphological Filtering: The original is a noisy binary image of “2” with isolated salt-noise speckles. And on the right is the output after morphological opening (disk, radius = 2) followed by morphological closing (disk, radius = 2). All speckle noise was removed and the digit shape was fully preserved.

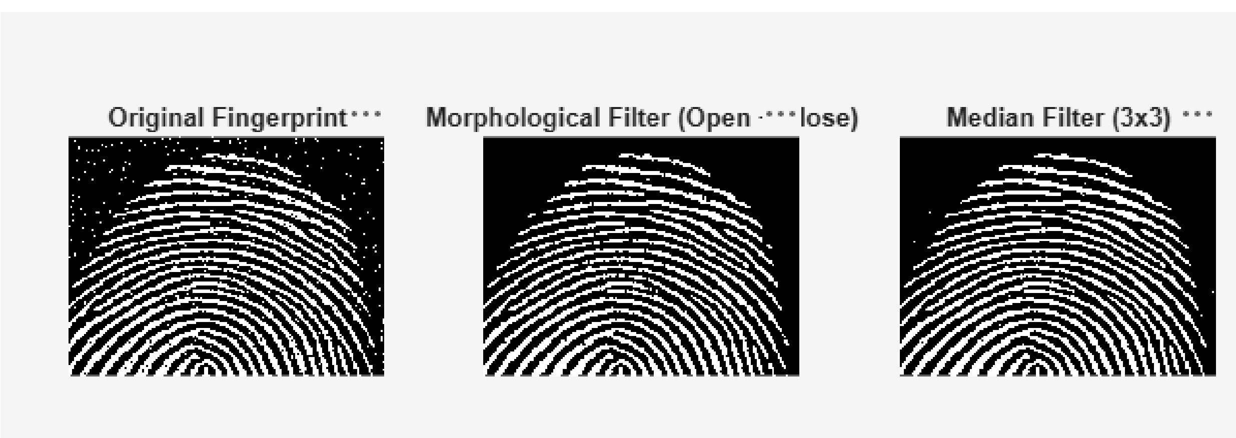

Fig2: Morphological Filter vs. Median Filter: First image is original binarized fingerprint image with speckle noise and the one next to it shows the output of the morphological filter (opening followed by closing, disk radius = 1).

And the last one shows the output of the median filter (3×3 window). Both filters removed the speckle noise while preserving the ridge pattern.

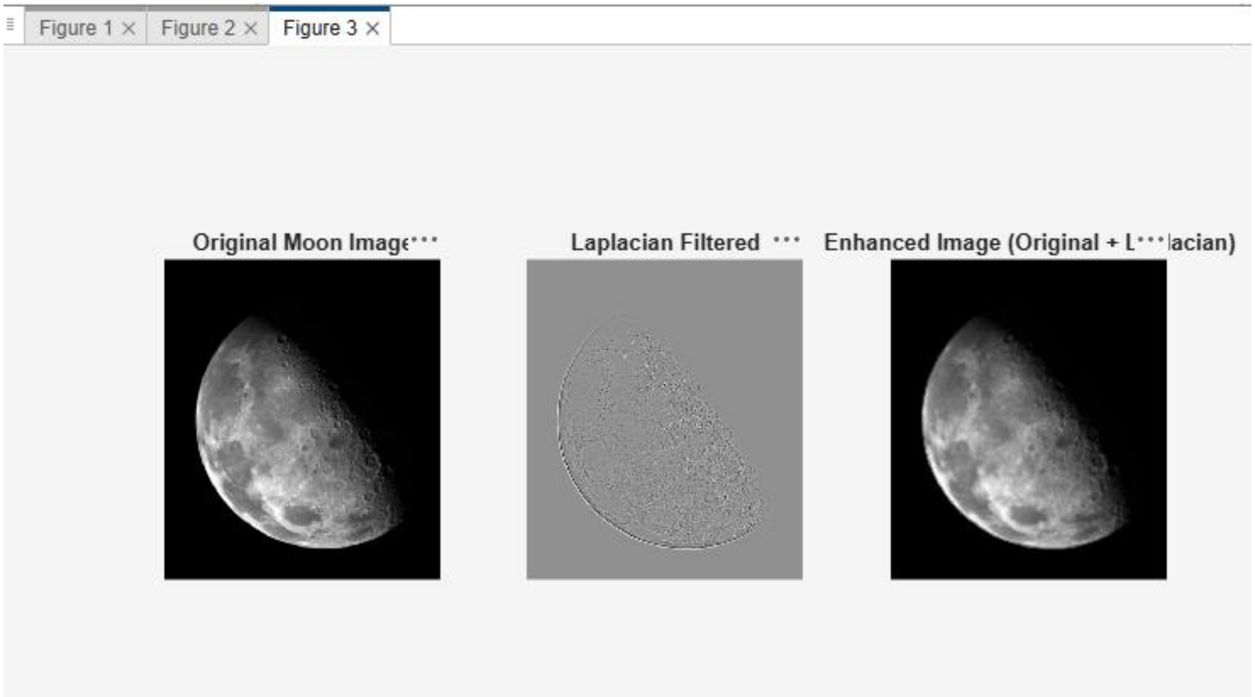

Fig3: Laplacian Filtering: First image is the original grayscale moon image and next one shows Laplacian-filtered response, highlighting crater rims, surface texture, and the lunar limb as bright edges against a mid-gray background and the last one shows the enhanced image produced by adding the Laplacian response to the original, showing visibly sharper crater edges and surface detail.

DISCUSSION:

This assignment helped me understand how different image filtering techniques work and how they improve image quality. Morphological filtering was effective for removing small objects and noise from binary images, while median filtering reduced noise while preserving important image details. The Laplacian filter enhanced edges and produced a sharper image by emphasizing intensity changes. Implementing these techniques in MATLAB gave me a better understanding of how different filters affect image processing results. One challenge was selecting the appropriate filter and parameters for each image. In the future, I would like to explore more advanced filtering techniques and compare their performance on different types of images.

CODE:

```
clc;
clear;
close all;

%% Morphological Filter
img = imread('morphology.png');
if size(img,3) == 3
    img = rgb2gray(img);
end

bw = imbinarize(img);
se = strel('disk', 2);

opened = imopen(bw, se);
cleaned = imclose(opened, se);
```

```
figure
subplot(1,2,1)
imshow(bw)
title('Input Image (Noisy "2")')

subplot(1,2,2)
imshow(cleaned)
title('Output Image (Opening + Closing)')

saveas(gcf, 'morphology_comparison.png');

%% Morphological Filter vs Median Filter
fp = imread('fingerprint_BW.png');
if size(fp,3) == 3
    fp = rgb2gray(fp);
end

fp_bw = imbinarize(fp);
se2 = strel('disk', 1);

fp_open = imopen(fp_bw, se2);
fp_morph = imclose(fp_open, se2);

fp_median = medfilt2(fp_bw, [3 3]);

figure
subplot(1,3,1)
imshow(fp_bw)
title('Original Fingerprint')

subplot(1,3,2)
imshow(fp_morph)
title('Morphological Filter (Open + Close)')

subplot(1,3,3)
imshow(fp_median)
title('Median Filter (3x3)')

saveas(gcf, 'fingerprint_comparison.png');
%% Laplacian Filtering on moon.jpg
moon = imread('moon.jpg');
if size(moon,3) == 3
    moon = rgb2gray(moon);
end

moon = im2double(moon);

H = fspecial('laplacian', 0.2);
lap = imfilter(moon, H, 'replicate');
enhanced = moon + lap;
enhanced = max(0, min(1, enhanced));

figure
subplot(1,3,1)
imshow(moon)
```

```
title('Original Moon Image')

subplot(1,3,2)
imshow(lap, [])
title('Laplacian Filtered')

subplot(1,3,3)
imshow(enhanced)
title('Enhanced Image (Original + Laplacian)')

saveas(gcf, 'moon_comparison.png');
```